



كيمياء عضوية

ك 217

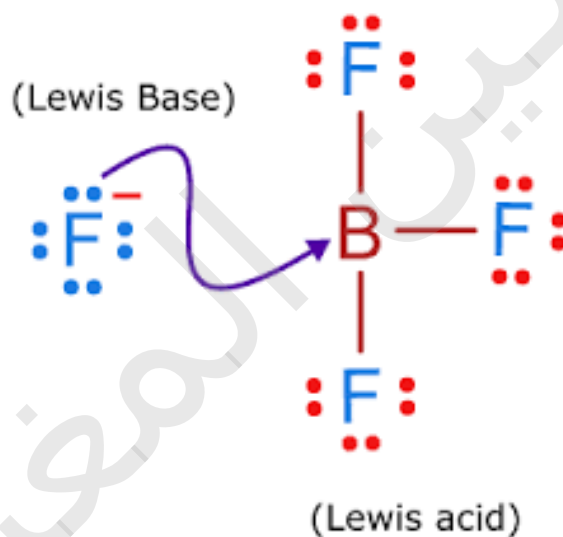
د. حسين المغيض

Chemistry Department

Organic Chemistry 217

Chapter 2

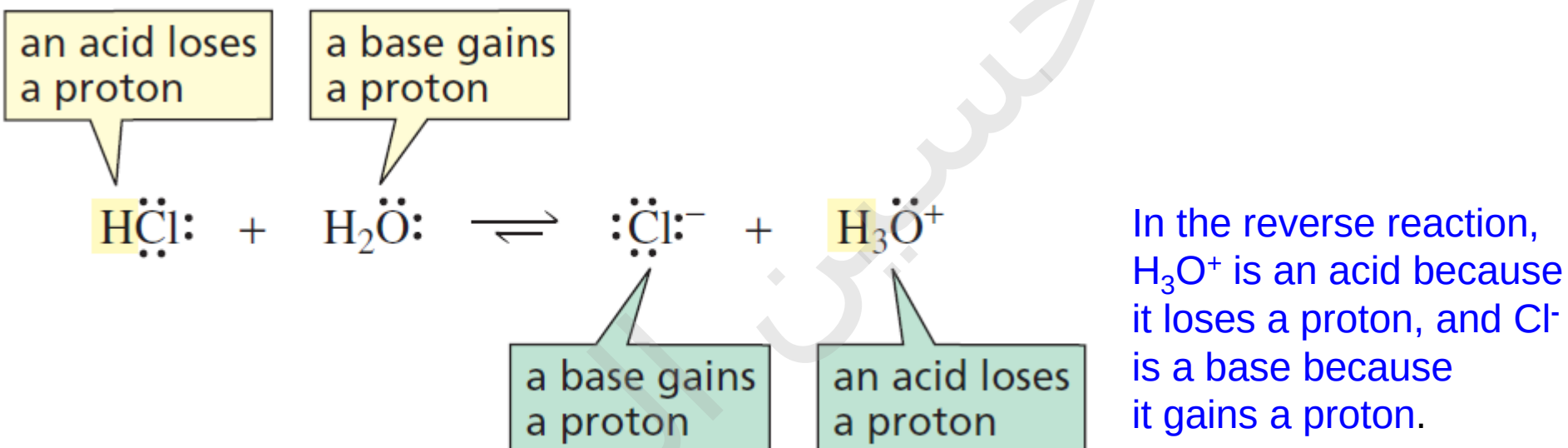
Acids and Bases



Dr. Hussein Al-Mughaid

Brønsted-Lowry Acid –Base Theory

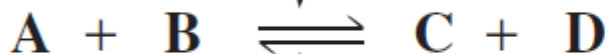
- acid = donates a proton (H^+)
- base = accepts a proton



*Any species that has a **hydrogen** can potentially act as an acid.
Any species that has a **lone pair** can potentially act as a base.*

Acid–base reactions

most acid–base reactions are reversible



a reversible reaction:

A and B form C and D.
C and D form A and B.



an irreversible reaction:

A and B form C and D, but
C and D do not form A and B.

an acid and its conjugate base



a base and its conjugate acid

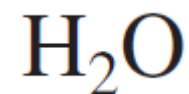
A conjugate base is formed by removing a proton from an acid.

A conjugate acid is formed by adding a proton to a base.

The stronger the acid, the weaker its conjugate base.

PROBLEM 3♦

a. What is the conjugate acid of each of the following?



b. What is the conjugate base of each of the following?



The degree to which an acid (HA) dissociates in an aqueous solution is indicated by the **acid dissociation constant**, K_a .



Brackets are used to indicate the concentrations of the reactants and products (in moles/liter).

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

The **larger** the acid dissociation constant, the **stronger** is the acid (**more readily loses a proton**).

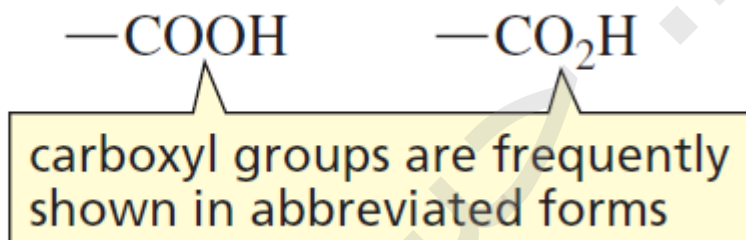
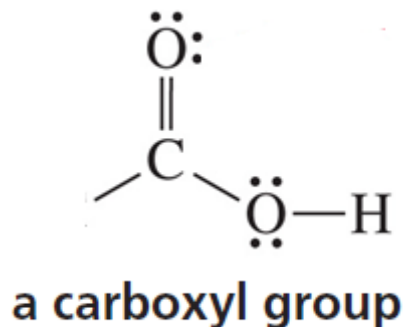
For convenience, the **strength of an acid** is generally indicated by its **$\text{p}K_a$** value rather than its K_a value.

$$\text{p}K_a = -\log K_a$$

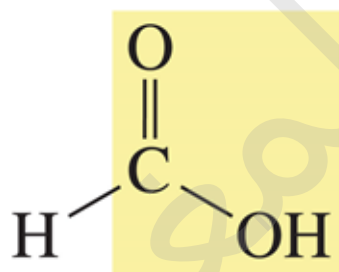
Notice that the stronger the acid, the smaller its $\text{p}K_a$ value.

Organic Acids

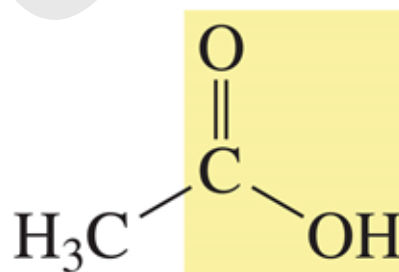
1. Carboxylic acids: compounds that have a carboxyl group



Weak acids: pK_a 3-5



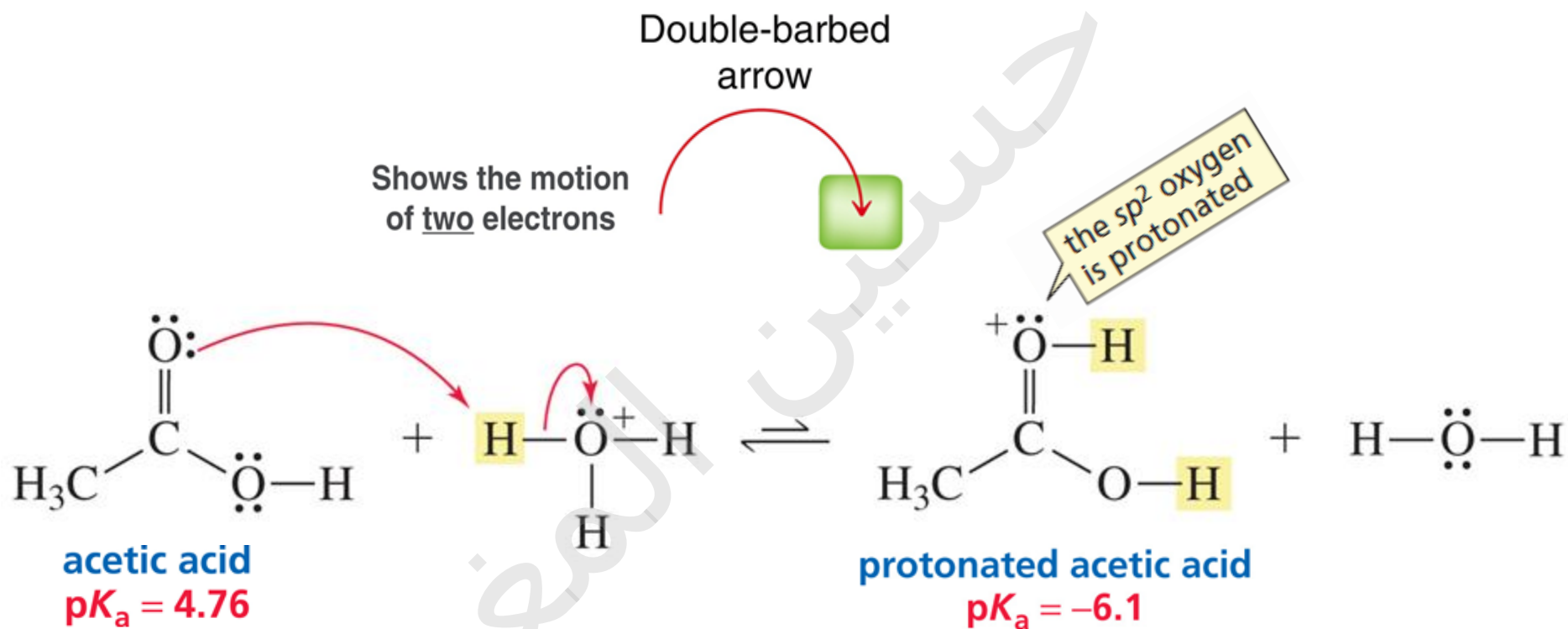
formic acid
 $pK_a = 3.75$



acetic acid
 $pK_a = 4.76$

Protonated carboxylic acids are very strong acids

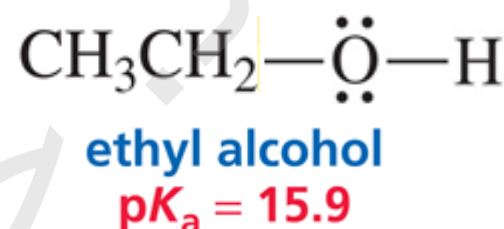
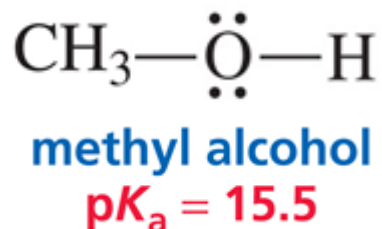
protonation = atom gain an additional H



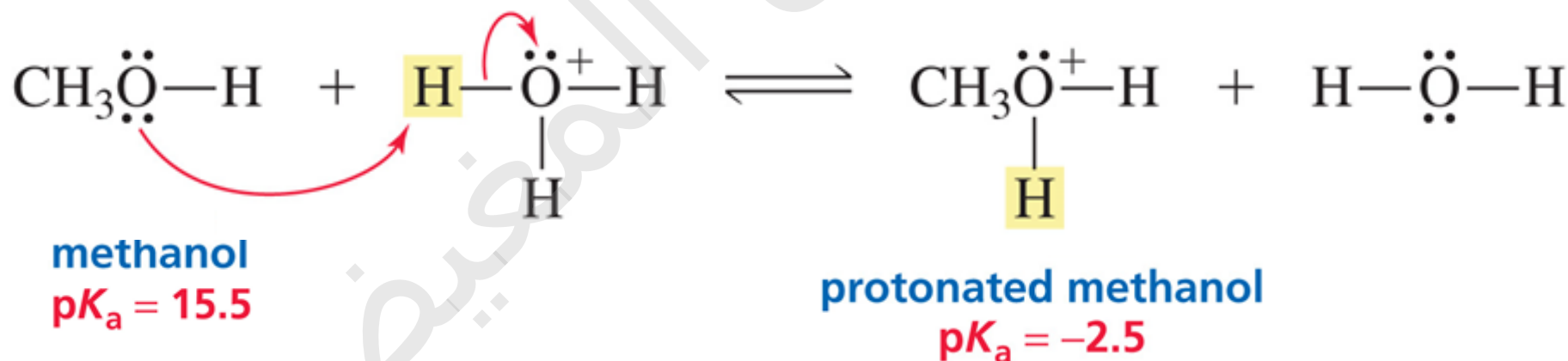
a lone pair of electrons from the sp^2 oxygen of the carboxylic acid takes a proton from a hydronium ion (note also the electrons from the bond returning as a lone pair to oxygen)

2. Alcohols: compounds that have an OH group

much weaker acids than carboxylic acids, with pK_a values close to 16.



 Protonated alcohols are very strong acids



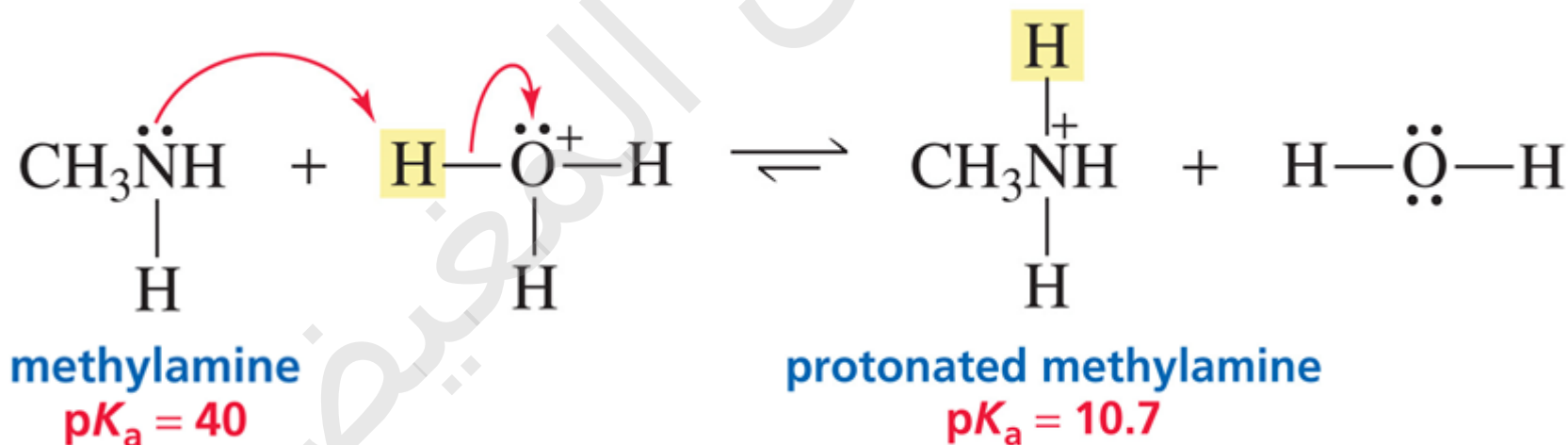
3. Amines: compounds that have a NH_2 group

Relatively weak acids (high $\text{p}K_a$) - more likely to act as a base

 **common organic bases**

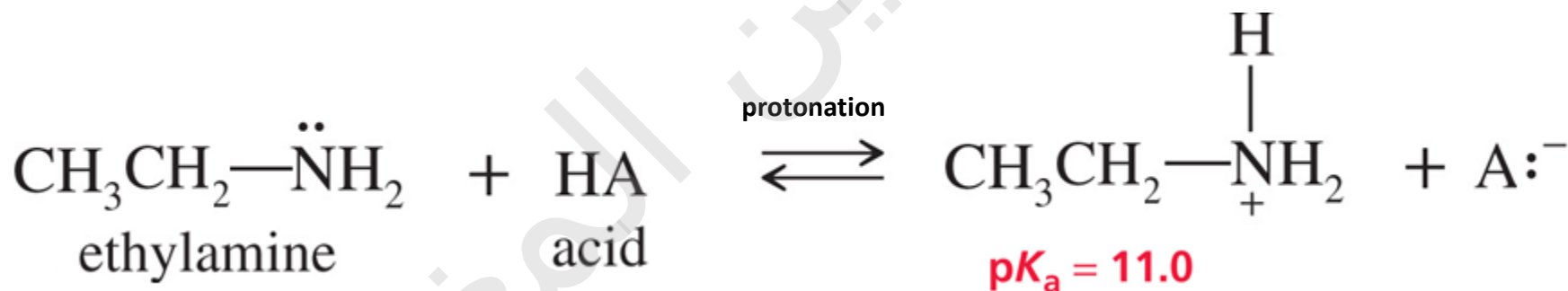
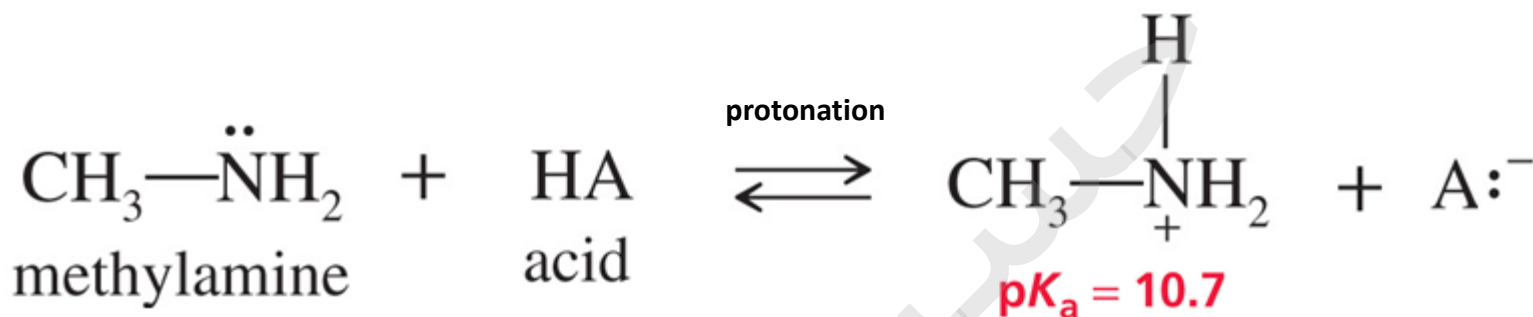
NH_3
ammonia
 $\text{p}K_a = 36$

CH_3NH_2
methanamine
 $\text{p}K_a = 40$



Thinking Question

☞ Which is a stronger base: methylamine or ethylamine ?



Since protonated methylamine is a stronger acid than protonated ethylamine, that must mean that methylamine is a stronger or weaker base than ethylamine?

It is **important** to know the **approximate pK_a values** of the various classes of compounds we have looked at so far.

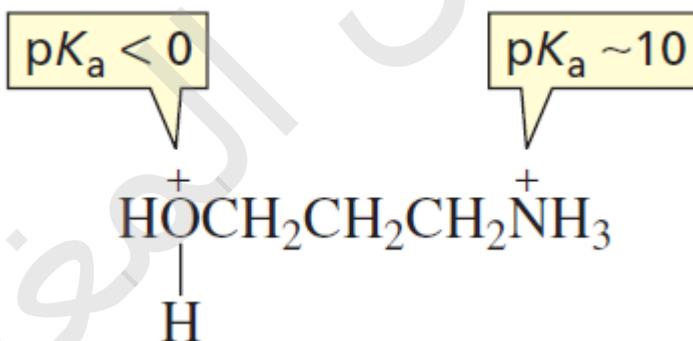


Table 2.1 Approximate pK_a Values

$pK_a < 0$	$pK_a \sim 5$	$pK_a \sim 10$	$pK_a \sim 15$
ROH_2^+ a protonated alcohol	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ a carboxylic acid	RNH_3^+ a protonated amine	ROH an alcohol
$\text{R}-\overset{+\text{OH}}{\parallel}{\text{C}}-\text{OH}$ a protonated carboxylic acid			H_2O water
H_3O^+ protonated water			

R is used when the particular carboxylic acid, alcohol, or amine is not specified.

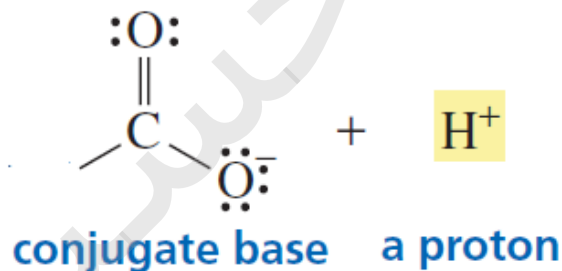
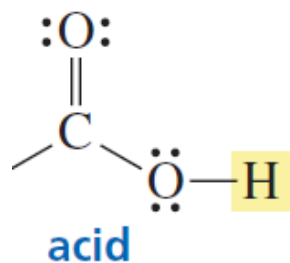
Question: Which atom of the following compound is more apt to be protonated?



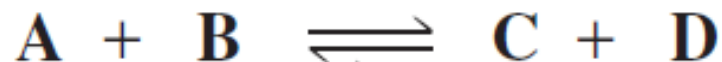
The conjugate acids have $\text{p}K_a$ values of ~ 0 and ~ 10 . Because the $^+\text{NH}_3$ group is the weaker acid, the NH_2 group is the stronger base, so it is the group more apt to be protonated.

PROBLEM 11♦

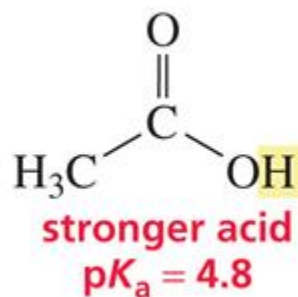
- a. Which is a stronger base, CH_3COO^- or HCOO^- ?
(The $\text{p}K_a$ of CH_3COOH is 4.8; the $\text{p}K_a$ of HCOOH is 3.8.)



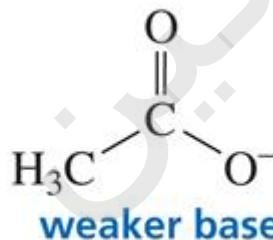
How do we determine the ***position*** of an acid-base equilibrium? Which favored reactants or products?



Compare the pK_a values of **reactant acid** and **product acid** molecules *on opposite sides of the equilibrium* (favors the formation of the weaker acid).



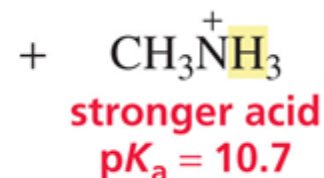
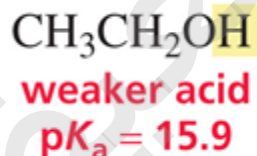
stronger base



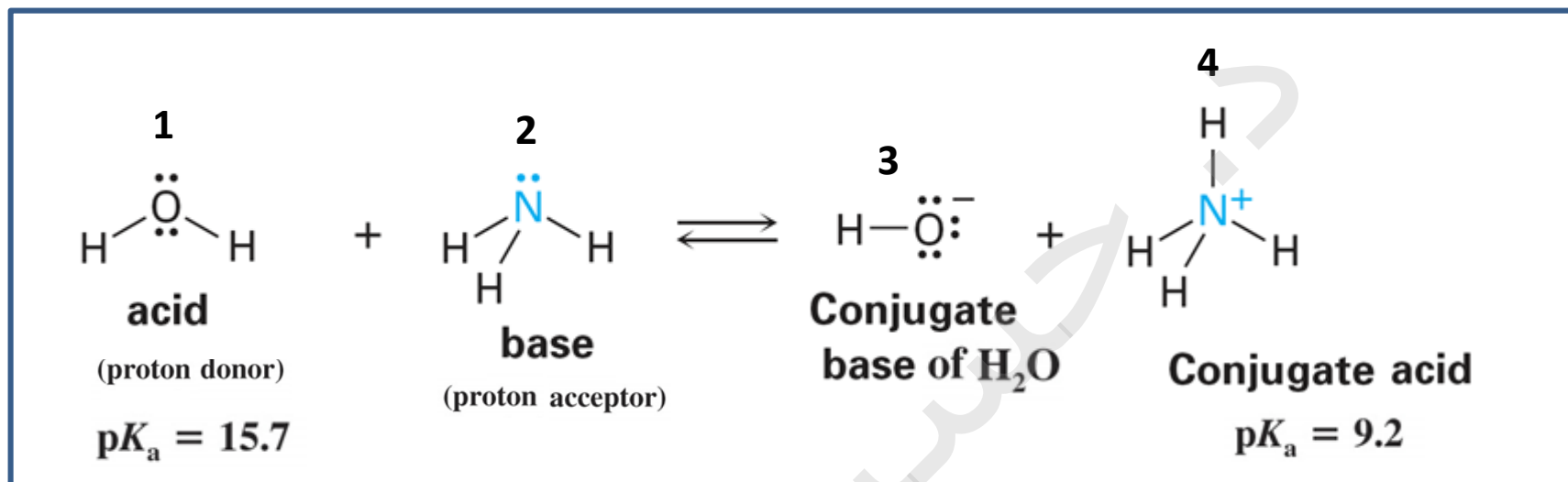
weaker acid
 $pK_a = 9.4$

products are favored because the weaker acid is a product

reactants are favored because the weaker acid is a reactant



For the following equilibrium:



1. Label Brønsted-Lowery acid(s) and Base(s)?

2. Label the acid-base conjugate pair(s)?

1 and 3

2 and 4

3. Does the equilibrium favor the reactants or the products)?

Higher $\text{p}K_a$ (weak acid) $\rightarrow$ reactants

How structure affects acidity

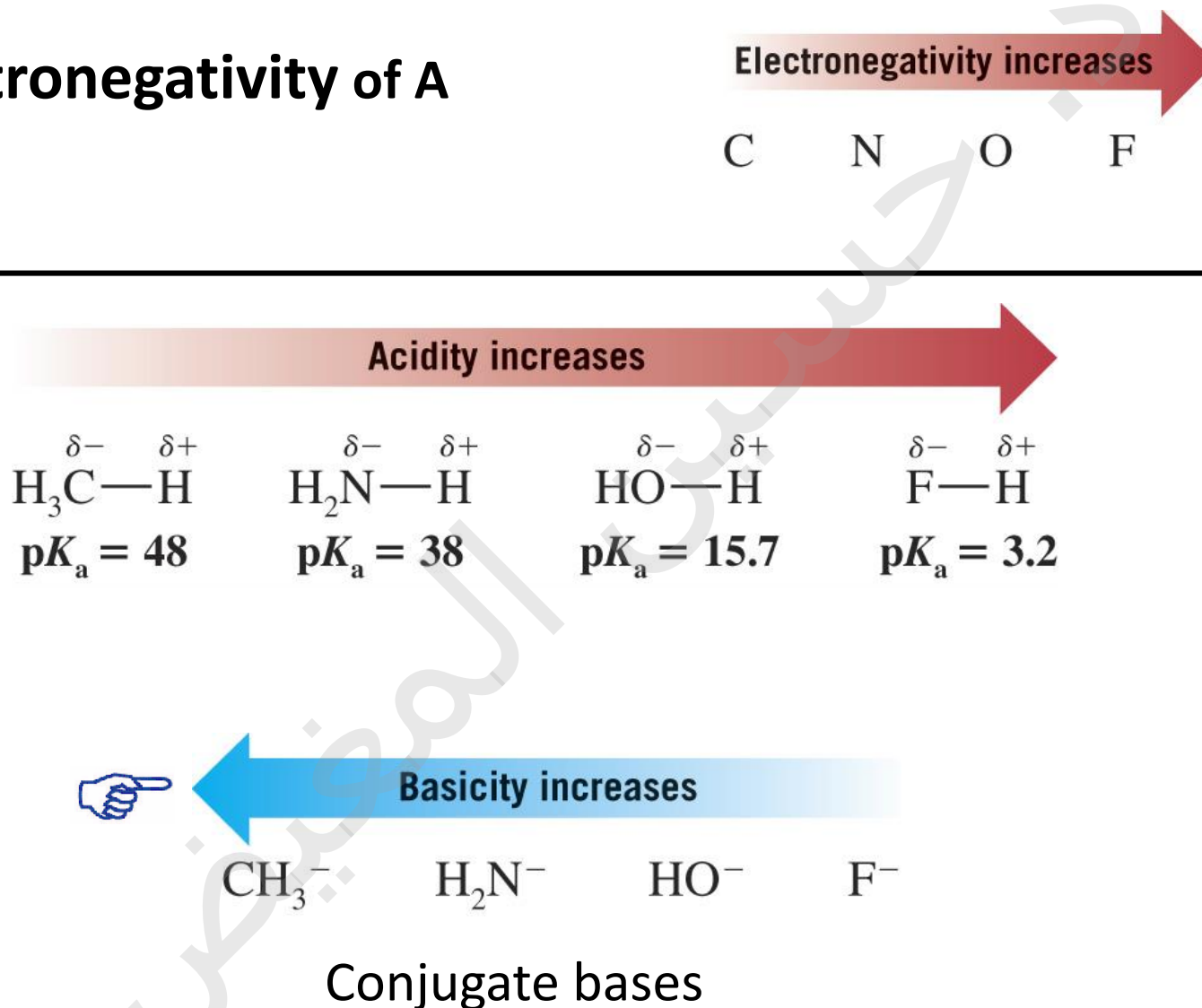
Acid strength: the extent of ionization of an acid,
measured by pK_a



➡ Any structural feature that stabilizes the conjugate base (A^-) with respect to its conjugate acid increases the acid strength by driving the equilibrium to the H_3O^+ and A^- side

Principal factors that affect the stability of A⁻

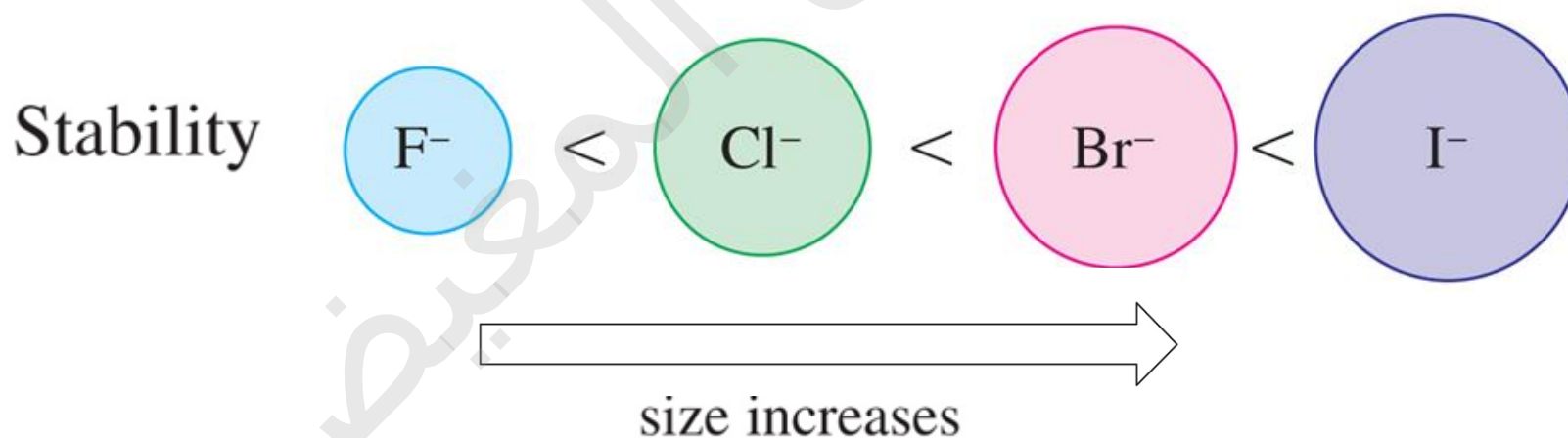
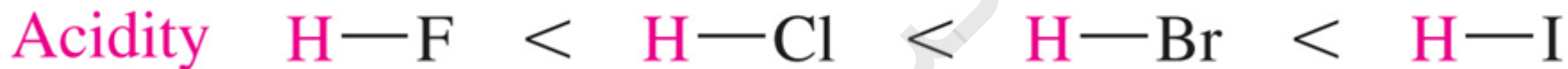
1. Electronegativity of A



2. Size of A

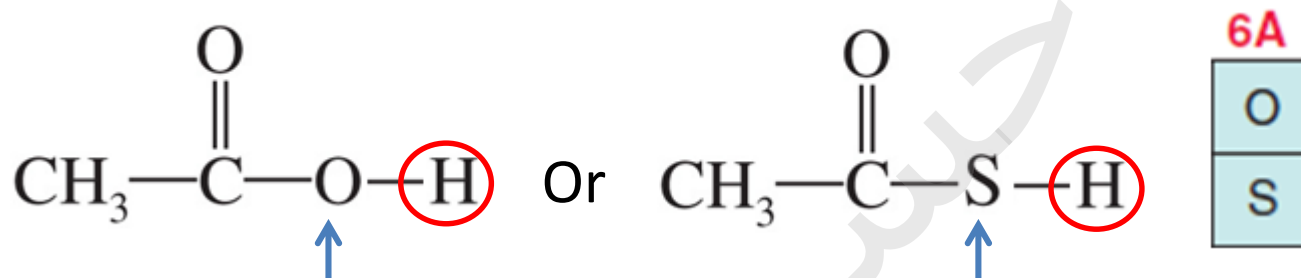
👉 As the size of A increases, the H is more loosely held and the bond is easier to break.

👉 A larger size also stabilizes the anion



Thinking Question

👉 Which compound would you expect to be more acidic?



👉 Electronegativity: $O > S$

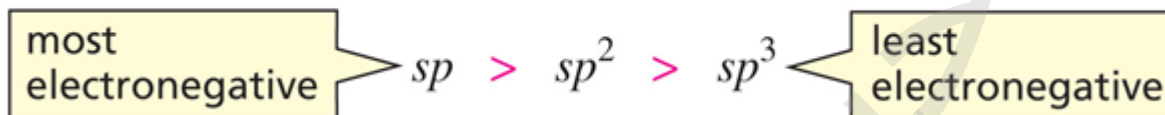
👉 Size: $S > 0$

3. Hybridization

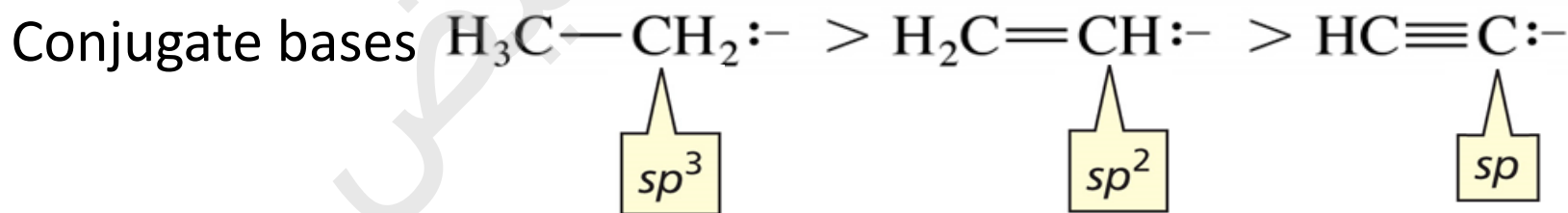
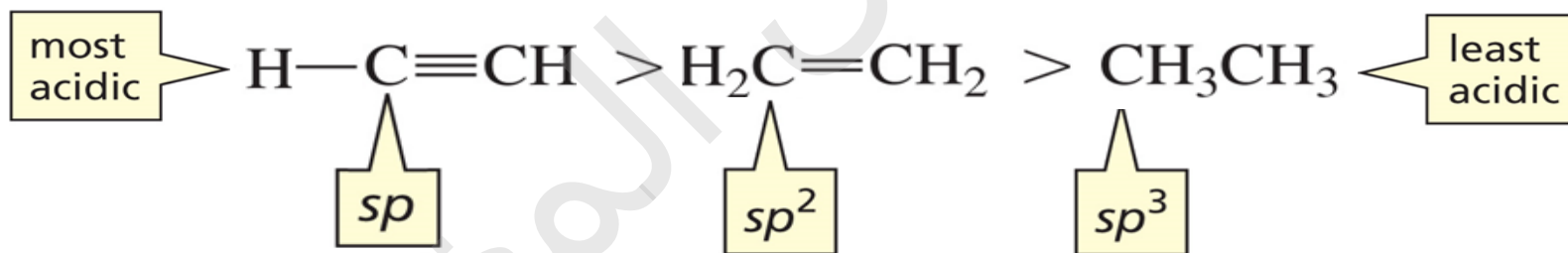


Hydrogens connected to orbitals with more s character will be more acidic

relative electronegativities of carbon atoms

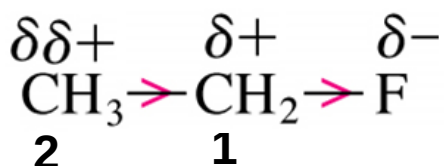


1. s orbitals are smaller and closer to the nucleus than p orbitals
2. Anions in hybrid orbitals with more s character will be held more closely to the nucleus and be more stabilized



How Substituents Affect Acidity

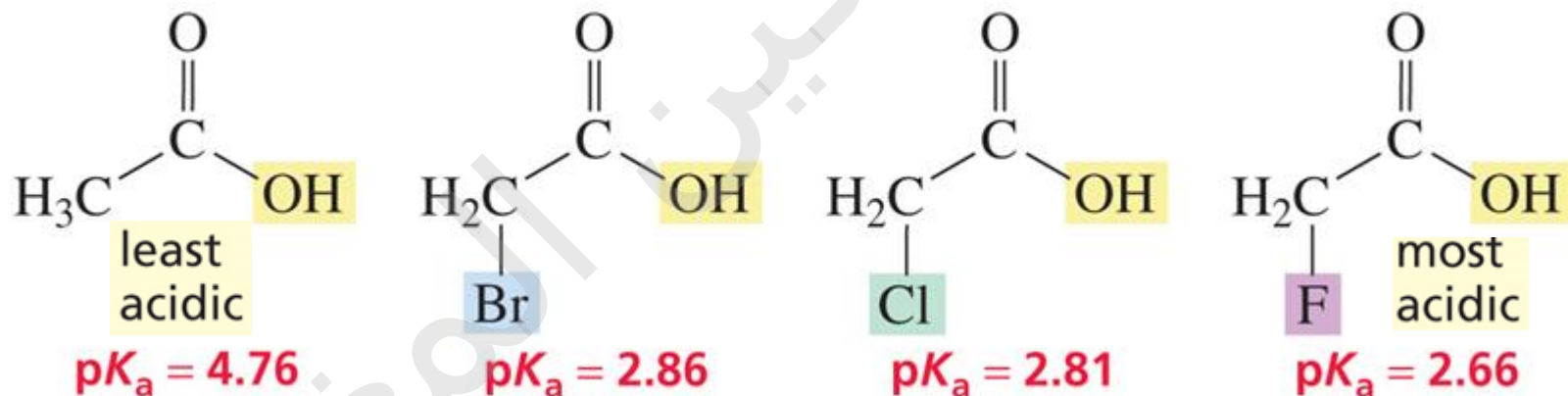
4. Inductive effect: Pulling electrons through sigma bond



the electronegative fluorine is withdrawing electron density away from the carbons
Fluorine is an electron withdrawing group (EWG)

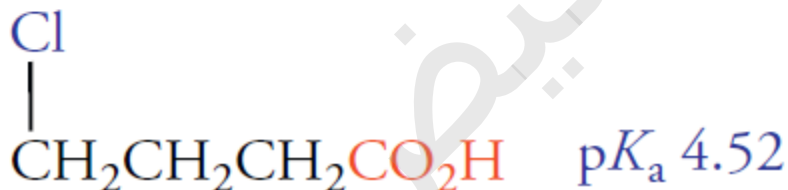
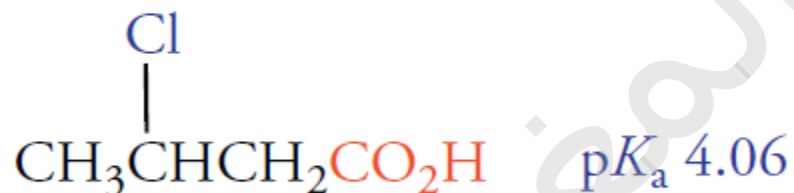
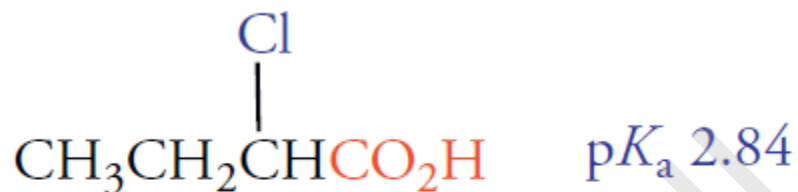
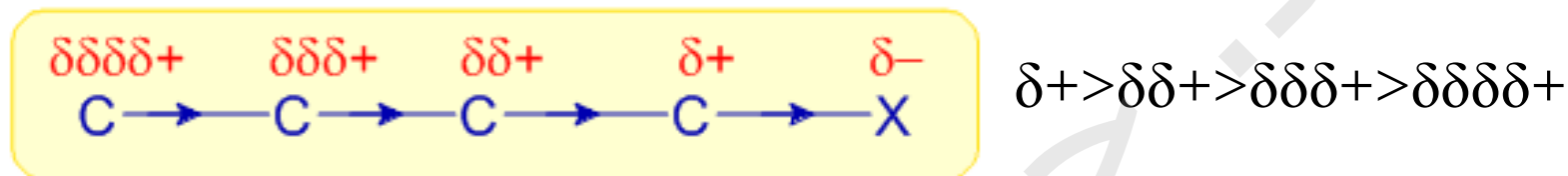
$\delta+ > \delta\delta+$

Case 1:



Case 2: distance effect

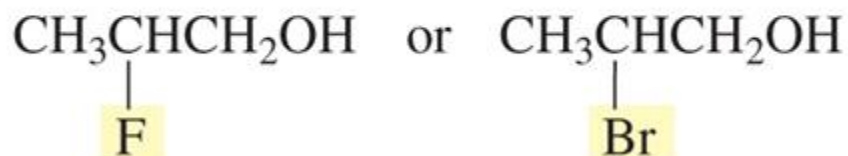
 Inductive effect gets weaker with increasing distance.



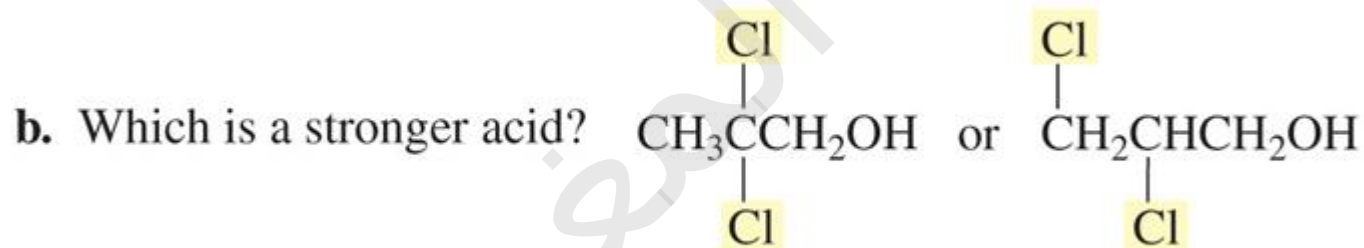
RULE: the effect of a substituent on the acidity of a compound decreases as the distance between the substituent and the acidic proton increases

Solved example

a. Which is a stronger acid?



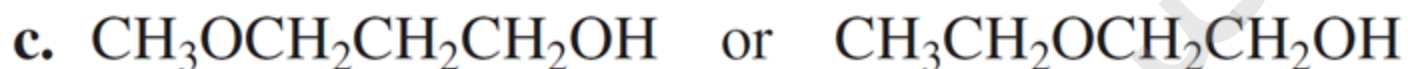
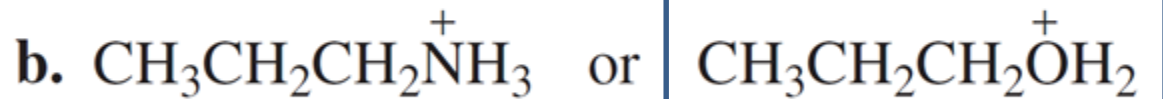
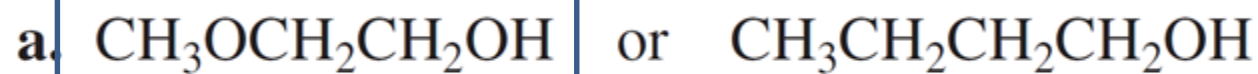
When you are asked to compare two items, pay attention to how they differ; ignore where they are the same. These two compounds differ only in the halogen atom attached to the middle carbon of the molecule. Because fluorine is more electronegative than bromine, there is greater electron withdrawal from the oxygen atom in the fluorinated compound. The fluorinated compound, therefore, will have the more stable conjugate base, so it will be the stronger acid.



These two compounds differ in the location of one of the chlorine atoms. Because the second chlorine in the compound on the left is closer to the O—H bond than is the chlorine in the compound on the right, the former is more effective at withdrawing electrons from the oxygen atom. Thus, the compound on the left will have the more stable conjugate base, so it will be the stronger acid.

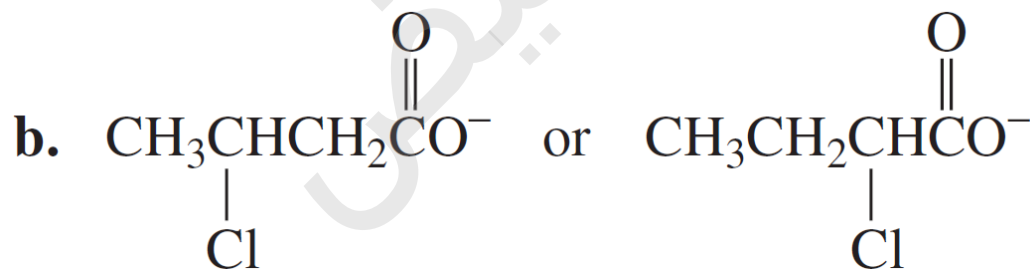
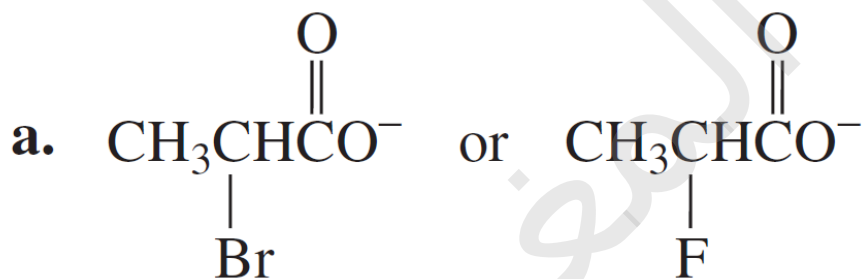
PROBLEM 26♦

Which is a stronger acid?



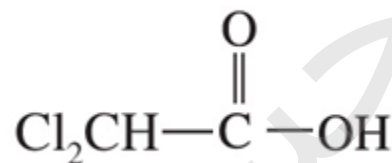
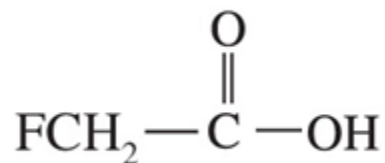
PROBLEM 27♦

Which is a stronger base?

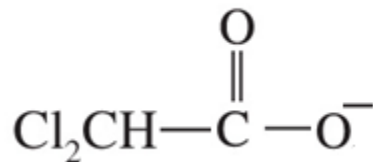
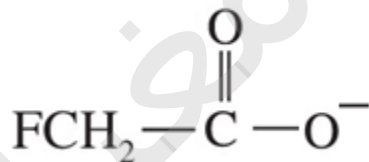


Thinking Questions

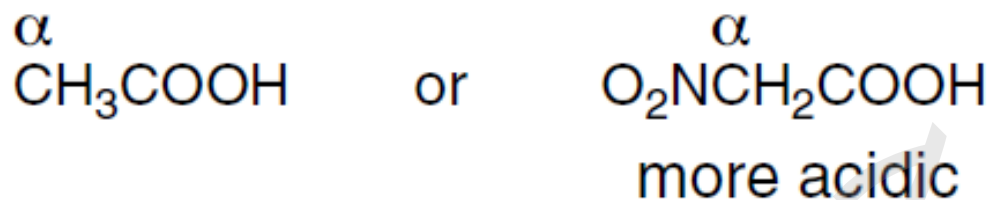
a. Which is a stronger acid?



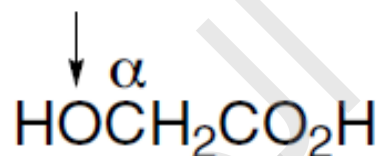
b. Which is a stronger base?



Other electronegative groups

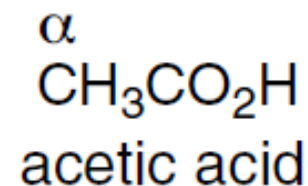


NO_2 is electron withdrawing, making the
O–H bond in the acid on the
right **more acidic**.



an α -hydroxy acid

or

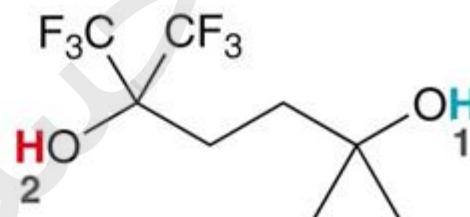


The extra OH group contains an electronegative
O, which stabilizes the conjugate base.
stronger acid

Why H2 is more acidic?

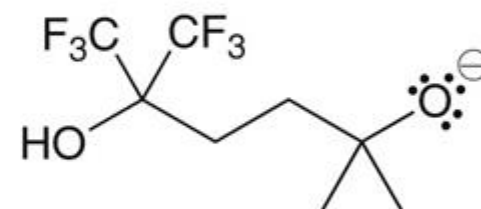
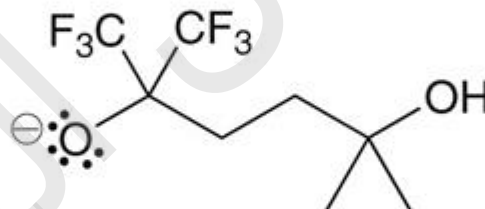


SOLVED



STEP 1

Draw the conjugate bases.

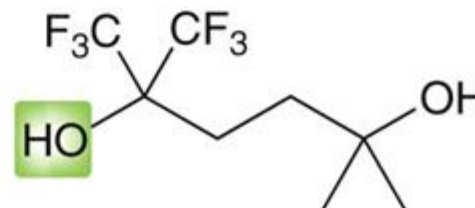


STEP 2

Look for inductive effects.

STEP 3

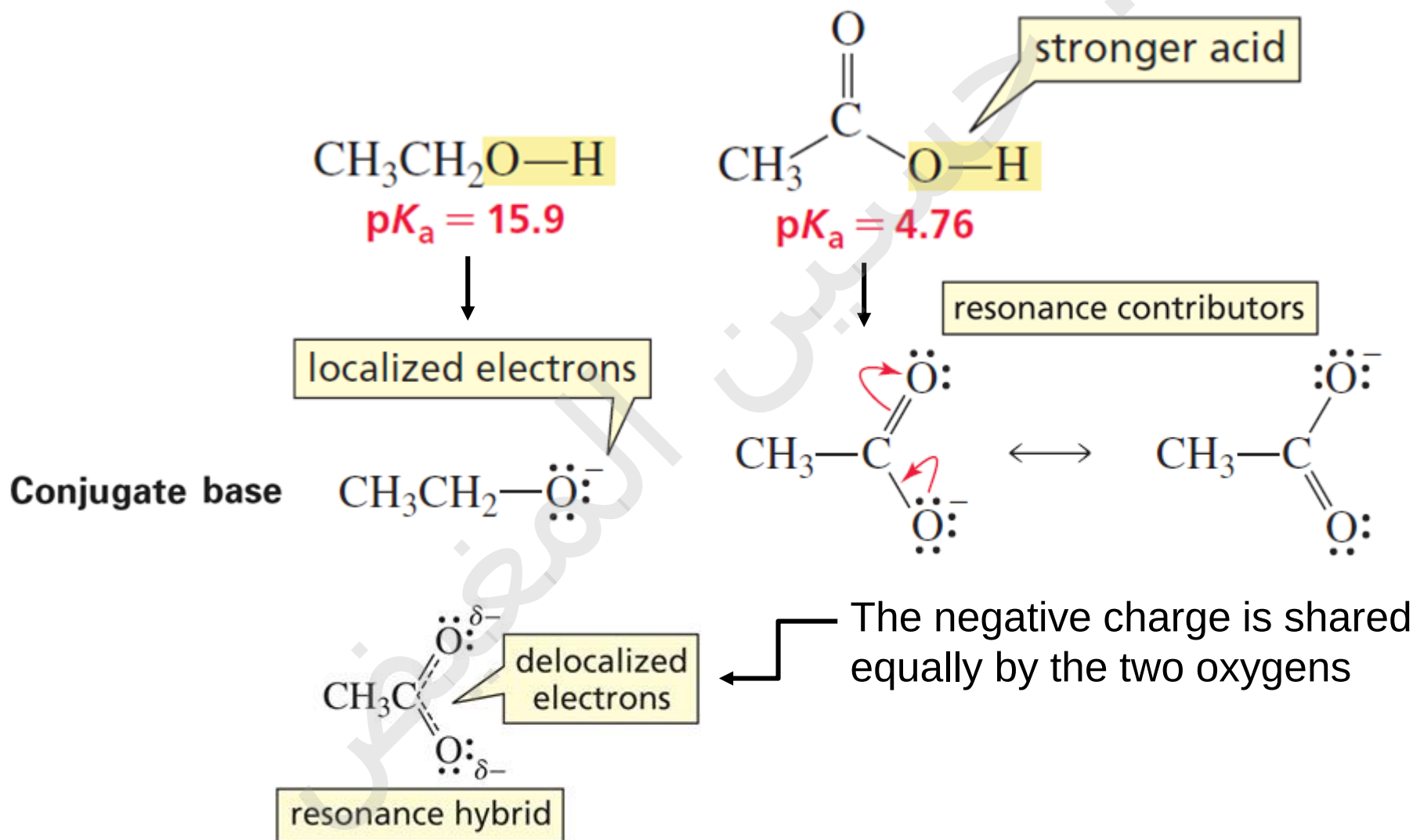
The more stable conjugate base corresponds with the more acidic proton.



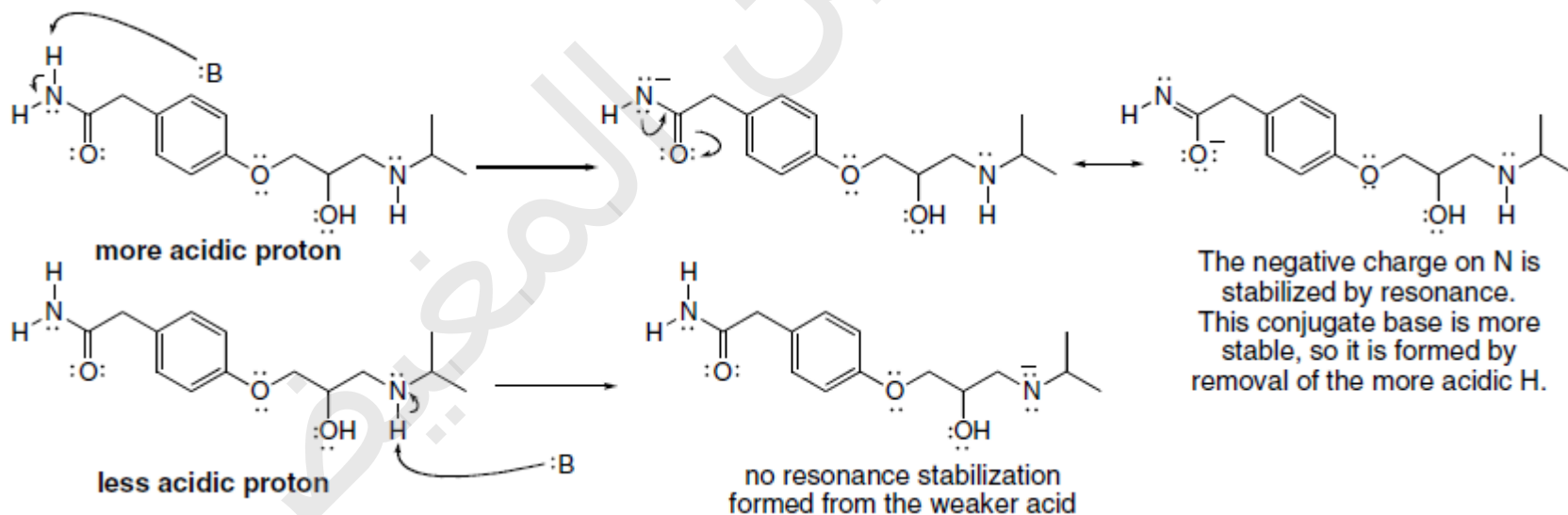
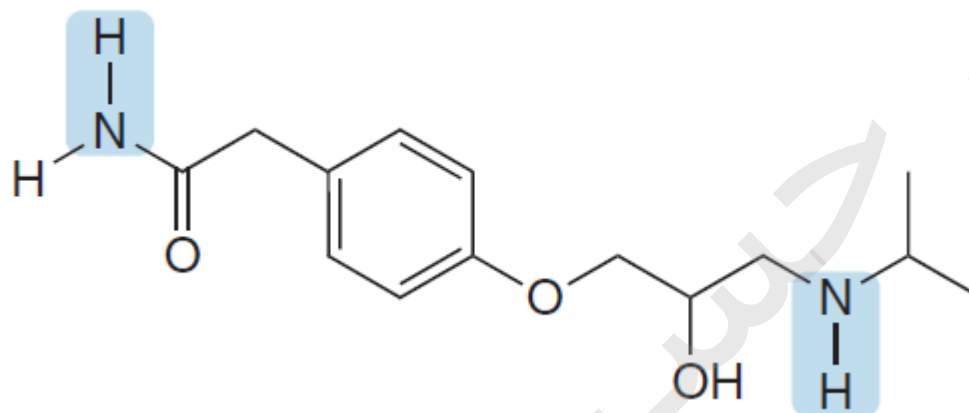
CF_3 is a strong electron withdrawing group

Acidity: Carboxylic acids Vs Alcohols

5. Resonance (Delocalized Electrons)



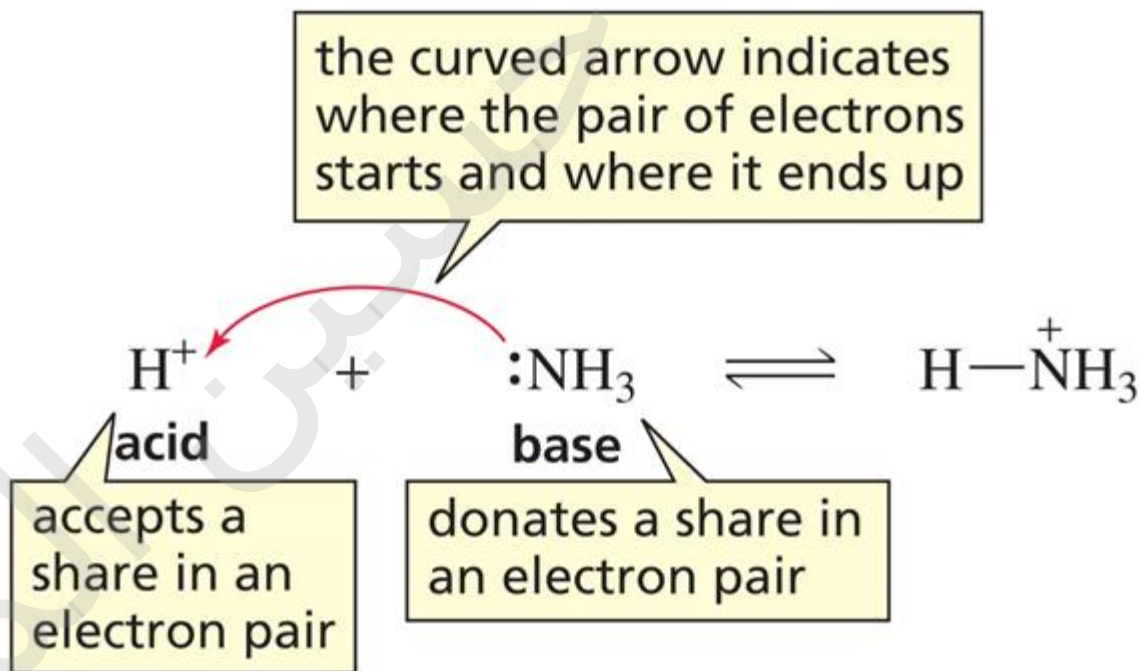
Which of the indicated N-H bonds is the most acidic



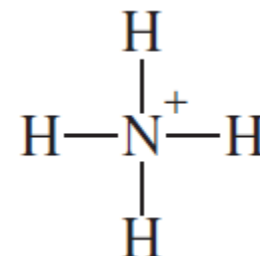
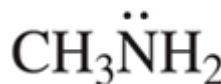
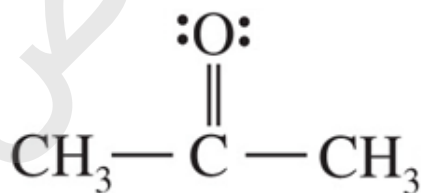
Lewis Acid –Base Theory

- A Lewis acid = an electron pair acceptor (non-proton-donating acid)
- A Lewis base = an electron pair donor.

NH_3 is classified as both a Lewis base (donates electron pair) and as a Brønsted base (accepts H)



Most oxygen- and nitrogen-containing organic compounds are Lewis bases because they have lone pairs of electrons.

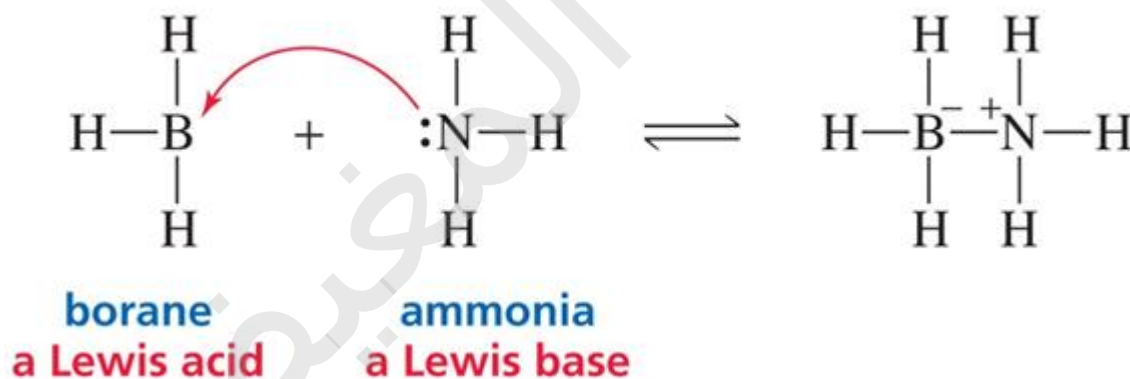
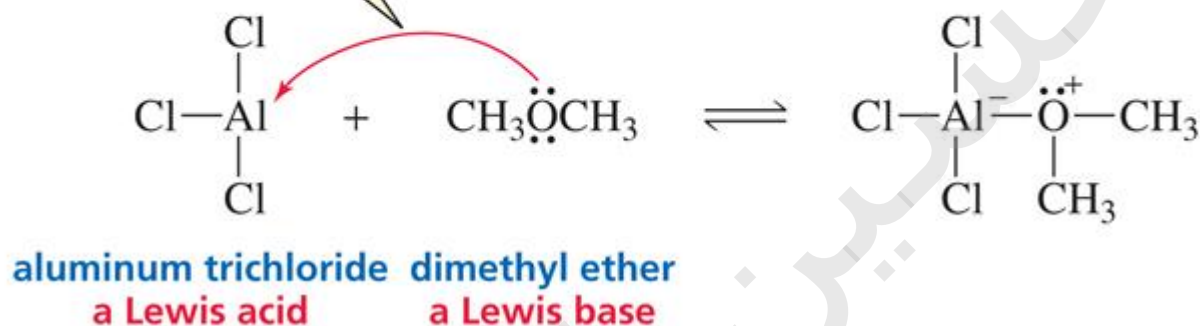




Group **3A** elements, such as **BF₃** and **AlCl₃**, **BH₃** are Lewis acids because they have unfilled valence orbitals and can accept electron pairs from Lewis bases.

3A
B
Al
Ga
In
Tl

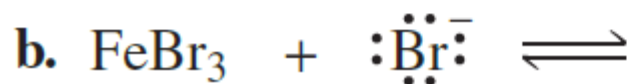
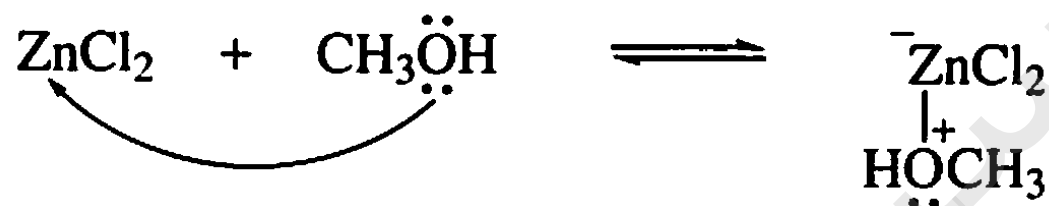
the curved arrow indicates where the pair of electrons starts and where it ends up



RULE: All Brønsted acids/bases are Lewis acids/bases, but not all Lewis acids/bases are Brønsted acids/bases.

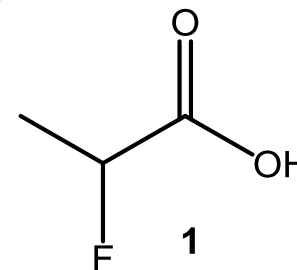
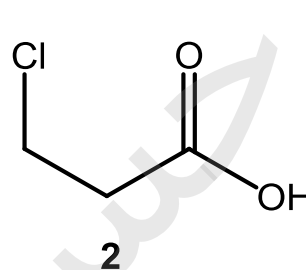
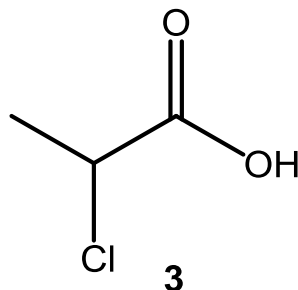
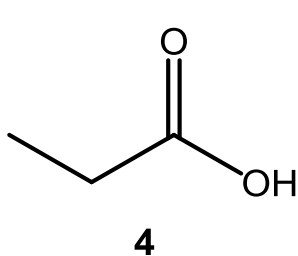
PROBLEM

Draw the products of the following reactions. Use curved arrows and where it ends up.



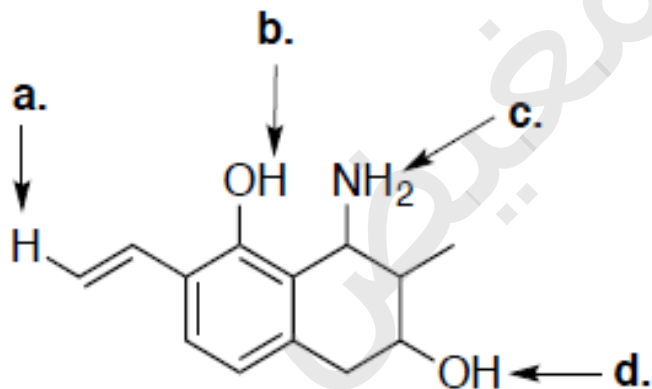
PRACTICE the skill

Q1) Rank the following carboxylic acids in order of decreasing acidity?



- A) $3 > 1 > 2 > 4$ B) $1 > 2 > 3 > 4$ C) $1 > 3 > 2 > 4$ D) $4 > 2 > 3 > 1$

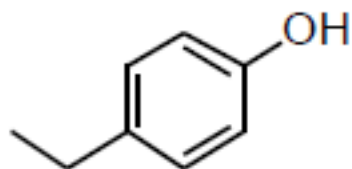
Q2) If the molecule below were reacted with a strong base, which proton would the base react with preferentially?



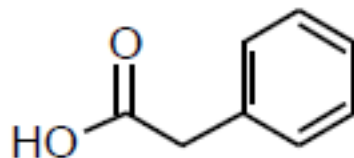
- A) a B) b C) c D) d

PRACTICE the skill

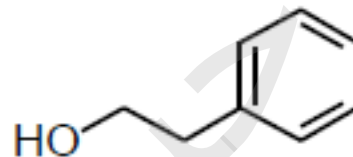
Q3) Rank the following compounds in order of increasing acidity?



A



B



C

1. $A < B < C$

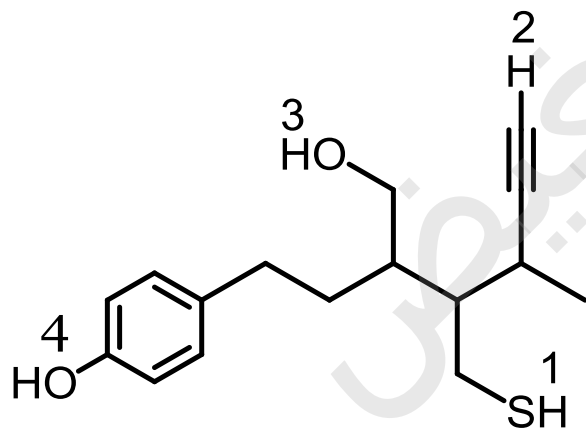
2. $B < C < A$

3. $B < A < C$

4. $A < C < B$

5. $C < A < B$

Q4) In the molecule shown below, which of the labeled hydrogens is the least acidic?



A) 1

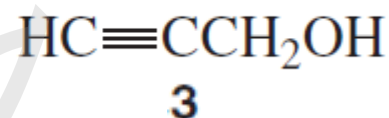
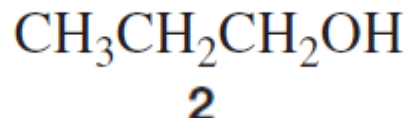
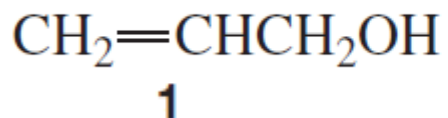
B) 2

C) 3

D) 4

PRACTICE the skill

Q5) Rank the following alcohols in order of decreasing acidity?



- A) $3 > 1 > 2$ B) $1 > 2 > 3$ C) $2 > 1 > 3$ D) $3 > 2 > 1$

Q6) Which compound/s can be deprotonated by OH^- , so that equilibrium favors the products? $\text{p}K_{\text{a}} \text{H}_2\text{O} = 15.7$

- | | | | |
|---|---|---|---|
| a. $\text{CH}_3\text{CO}_2\text{H}$
$\text{p}K_{\text{a}} = 4.8$ | b. $\text{CH}_3\text{CH}_2\text{NH}_2$
$\text{p}K_{\text{a}} = 40$ | c. $\text{CH}_3\text{CH}_2\text{C}\equiv\text{C}-\text{H}$
$\text{p}K_{\text{a}} = 25$ | d. $\text{CH}_3\text{CH}_2\text{NH}_3^+$
$\text{p}K_{\text{a}} = 10.7$ |
|---|---|---|---|

- A) only a B) a and d only C) b and c only D) only d